Technical Supplement
Towards Radar-Aware VLM-Based Autonomous Driving via mmWave Radar-Vision-Language Reasoning

Anonymous submission

This document is the technical supplement to the main paper. The main paper is self-contained; this supplement is not required reading and only provides additional background, benchmark-construction details, prompt templates, method derivations, and extended experiments that support the claims in the main paper. References of the form "Appendix X" in the main paper correspond to the lettered sections (A–J) below. To avoid redundancy, result tables that already appear in the main paper are not repeated here; this supplement reports only the construction details, protocols, and additional analyses behind them. Appendix K lists every evaluation subset used in this paper together with its definition and size.

Ethical Considerations

Potential risks. Our work targets the evaluation of radar-vision-language reasoning in decision-critical driving scenarios and does not directly produce deployable driving policies. Nonetheless, we acknowledge that improvements in radar-aware autonomous driving systems carry dual-use risks if deployed without adequate human oversight, particularly in safety-critical settings. We advocate for responsible deployment, rigorous validation, and the retention of human-in-the-loop supervision for any system informed by our findings.

Independent human verification. The gold answers in mmDrive-QA are conditioned on physical quantities (speed, acceleration, distance, TTC) computed from the source sensor metadata, including ego-motion (IMU) signals, object tracks, and LiDAR 3D boxes, rather than freely generated from video content alone. The verifier's role is therefore not a subjective judgment but a consistency check of whether the gold answer agrees with these physical quantities, which structurally limits annotator bias. To ensure independence, all human verification of QA pairs (answer balancing, question refinement, and validity checks described in Appendix C) was conducted by three annotators who are not authors of this work. Each reviewed the complete set of 2,019 QA pairs, receiving the video together with the question, options, and gold answer, and flagged any item whose gold answer was inconsistent with the visual or sensor evidence, leaked the correct option, or contained any other defect (Table 1). The three annotators flagged 3, 3, and 15 items; we jointly adjudicated the union of 13 flagged items and confirmed that none constitutes an actual error. This review is a post-hoc audit of the *final* 2,019 QAs and is distinct from the construction-time verification that discarded 221 candidates (Appendix C.2); the zero-error outcome is therefore a property of the retained set, not a claim that generation was error-free. No crowdworkers were recruited, and therefore no external compensation or recruitment protocol was required.

Metric	Result
Annotators (non-authors)	3
QA pairs reviewed (each, full set)	2,019
Items flagged (per annotator)	3 / 3 / 15
Flagged by $\geq$1 annotator (union)	13 (0.64%)
Flagged by $\geq$2 annotators	2 (0.10%)
Confirmed actual errors	0

Table 1: Independent review of all 2,019 QA pairs by three non-author annotators. Adjudication of the union of flagged items confirmed no actual errors.

Use of AI assistants. We use Gemini 2.5 Flash to generate candidate QA pairs from selected video clips (Appendix G.1) and GPT-5-mini as a text-only probe for distractor-leakage detection during the hardening procedure (Appendix C.3). All AI-generated content was subsequently verified by human annotators. AI assistants were also used for light editing of the manuscript text; all technical content, experimental design, and analysis are the authors' own work.

A Autonomous Driving Systems

Autonomous driving systems (ADSs) have become a key technology for modern intelligent transportation, enabling vehicles to perceive surrounding environments, predict future traffic behaviors, and plan safe driving actions (Yurtsever et al. 2020). Traditional ADSs are primarily built upon data-driven pipelines and are commonly categorized into modular and end-to-end frameworks (Tampuu et al. 2022; Chen et al. 2024).

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Modular ADSs decompose the overall driving process into several specialized components, including perception, prediction, and planning. The perception module extracts information from the surrounding environment and identifies important traffic elements such as vehicles, pedestrians, lanes, and obstacles. Representative perception tasks include 3D object detection and occupancy prediction, which provide structured scene representations for downstream decision-making (Wang et al. 2021; Sima et al. 2023). The prediction module estimates the future motion of surrounding traffic participants based on the perceived scene, typically through trajectory or motion forecasting (Gao et al. 2020; Shi et al. 2022). The planning module then generates safe and comfortable driving routes or actions by considering the outputs of perception and prediction (Chen et al. 2020). Although modular systems have achieved remarkable progress across many driving scenarios, their cascaded design may cause information loss during inter-module transmission, introduce redundant computation, and accumulate errors across stages. Moreover, since each module is often optimized with different objectives, local improvements in individual components do not necessarily lead to globally optimal driving decisions (Hwang et al. 2025).

In contrast, end-to-end ADSs integrate the driving pipeline into a unified neural network that directly maps sensor observations to future trajectories or control actions (Pomerleau 1988). By jointly optimizing perception, prediction, and planning, end-to-end approaches can reduce hand-crafted interfaces between modules and alleviate error propagation. Recent studies have explored end-to-end driving with camera, LiDAR, BEV representations, and unified planning-oriented architectures (Chitta et al. 2023; Hu et al. 2022, 2023). However, these models often suffer from limited interpretability, as their decision-making process is embedded in a complex neural network. This "black-box" nature makes it difficult to verify why a particular action is selected, which remains a critical limitation for safety-critical autonomous driving (Shao et al. 2023).

B Benchmark Curation

This appendix consolidates the construction details of mmDrive-QA Bench: video selection (video selection subsection), the radar representation used across all evaluation conditions (radar representation subsection), and the radar text format consumed by the VLM (radar text format subsection).

B.1 Video Selection

We build our benchmark on top of the **NVIDIA PhysicalAI-Autonomous-Vehicles** dataset,[1] which provides synchronized multi-camera images, LiDAR point clouds, imaging radar, and per-frame egomotion data. The full corpus contains 74.6K clips of 20-second duration. Since the corpus is dominated by normal driving scenarios, we curate a focused subset of **140 edge-case clips** that stress different aspects of vision-only autonomous driving systems.

[1]https://huggingface.co/datasets/nvidia/PhysicalAI-Autonomous-Vehicles

Data source and licensing. mmDrive-QA is built on the publicly released NVIDIA PhysicalAI-Autonomous-Vehicles dataset under its license. We release only our QA annotations (questions, options, gold answers, rationales, taxonomy labels, and clip UUIDs) under the CC BY-NC-SA 4.0 license, without redistributing the source sensor data. Researchers wishing to reproduce our experiments can obtain the underlying sensor data directly from NVIDIA under its respective license terms.

Edge-case taxonomy. Drawing from prior characterizations of failure modes in vision-based driving benchmarks, we organize edge cases along four behavioral axes: *unstructured/OOD agents*, *spatial complexity*, *dynamic interaction*, and *sensory degradation*. We decompose these axes into eight scenario categories: Micro-Mobility (Wilson et al. 2021), Complex Intersection Crossing (Zhan et al. 2019), Roundabout (Li et al. 2022), Occlusion & Dart-out, Aggressive Cut-in, Sudden Braking (Wilson et al. 2021), Entry/Exit, and Adverse Weather & Low-light.

Two-stage automatic extraction. We design a two-stage pipeline that scans the entire 74.6K-clip corpus to extract clips matching the eight categories above (see the mmDrive-QA curation pipeline figure in the main paper).

Stage A: Kinematic and object-tracking filter. Using the per-clip egomotion parquet (interpolated to 30 fps) and per-frame object tracks obtained with YOLO11, we flag clips exhibiting kinematic or interaction signatures characteristic of each category. We threshold on longitudinal deceleration ($a_x < -3.0\,\mathrm{m/s^2}$), lateral acceleration ($|a_y| > 1.5\,\mathrm{m/s^2}$), cut-in IoU ($> 0.10$), and category-specific track behaviors. Each event must persist over a 30-frame (1-second) window to suppress single-frame noise, and tracks shorter than 5 frames are discarded as detection artifacts.

Stage B: VLM-based category verification and severity scoring. Stage-A candidates are passed to **Gemini 2.5 Flash**, which performs blind 8-way classification (plus a *none* bucket) and assigns a semantic severity score on a 1–10 scale. Within each category we retain the top-ranked clips by severity until we reach 140 clips, balanced across the eight categories (16–20 clips each).

B.2 Radar Representation for VLMs

Existing VLMs accept only images and text. To make radar point clouds consumable by such interfaces without retraining, we serialize each radar frame as a fixed-format text block. The format is designed so that it mirrors how VLMs already consume multi-frame visual inputs, Doppler sign encodes motion direction in a human-readable convention, and the per-frame point budget is bounded to prevent token explosion. The same representation is used uniformly across all evaluation conditions.

Reference frame and Doppler convention. All radar points are expressed in the ego-vehicle rig frame ($+x$ forward, $+y$ left, $+z$ up). Doppler velocity is ego-motion compensated: a negative Doppler indicates closing on the ego vehicle, while a positive Doppler indicates separation.

Per-frame cap rule. A single 20-second clip may contain tens of thousands of radar returns. To bound token cost, each frame is capped at $M \leq 200$ points using the rule: retain the union of (a) the top-100 points by absolute Doppler $|v_d|$ and (b) the top-100 points by intensity. This is exactly the $\mathtt{R}_{\mathtt{cap}}$ baseline used in the main paper. The cap reduces per-QA token length by approximately $5\times$, yielding a median of $\sim$40K tokens per QA over the full 20-frame sequence under the GPT tokenizer. Token counts differ substantially across tokenizer families because radar text is dominated by floating-point literals, which different tokenizers segment very differently; under the Qwen tokenizer the same radar block averages $\approx$108K tokens. Measured per-condition counts are reported in Appendix I.9.

B.3 Radar Text Format

Each radar frame is emitted as a fixed-header block followed by a tabular point listing. For frame N at time $t = (N-1)$ s, the block contains the ego velocity (v_x, v_y, v_z) followed by up to 200 scene points, each as a row of $(x, y, z, \text{doppler}, \text{intensity})$. The resulting text fed to the VLM takes the form:

Radar Text Format Example

```
========== [ FRAME 1 | t = 0.0 s ]
    ==========
[Ego Velocity] vx=12.43, vy=-0.18, vz
    =0.00 (m/s)
[Scene points] (M = 178)
      x y z doppler intensity
  18.42 -2.15 0.31 -3.12 52.4
  24.07 1.88 -0.05 -2.94 48.1
  31.55 -0.42 0.12 -2.71 45.7
   ...
========== [ FRAME 2 | t = 1.0 s ]
    ==========
[Ego Velocity] vx=12.10, vy=-0.21, vz
    =0.00 (m/s)
[Scene points] (M = 184)
      x y z doppler intensity
  ...
```

A negative Doppler value (e.g., -3.12 m/s) indicates the point is approaching the ego vehicle. The same fixed format is used across all frames per QA and identically across all radar-input conditions.

C Question Generation and Verification

This appendix consolidates the QA construction and verification details: question generation and hardening (question generation subsection), final benchmark statistics (statistics subsection), detailed hardening procedures (hardening subsection), and the numeric-leakage audit of the radar text (audit subsection).

C.1 Question Generation and Hardening

For each of the 140 selected clips, we generate questions via **Gemini 2.5 Flash** with a fixed master prompt (see Appendix G.1). Each clip yields exactly 16 raw questions, four per reasoning taxonomy (2,240 candidate QAs total). The four taxonomies are designed to probe complementary aspects of driving-scene understanding. *Perception* covers the state, exact location, depth, relative velocity, or occlusion of target objects in the scene. *Anticipation* covers the future trajectory, speed change, or time-to-collision (TTC) of the ego vehicle or surrounding objects. *Reasoning* covers the physical or interactive cause underlying the critical event in the scene. Finally, *Planning* covers the safest immediate action the ego vehicle should take given the environment and physical constraints. Each question is at most 50 words with five options of at most 15 words each, a gold letter, and a free-form rationale. Appendix J shows eight representative items, one per scene category.

Reasoning Taxonomy	# QAs
Perception	498
Anticipation	512
Reasoning	493
Planning	516
Total	**2,019**

Table 2: Final QA distribution across the four reasoning taxonomies after hardening and human verification.

A common failure mode of VLM-generated benchmarks is that questions become solvable from text alone. We mitigate this through a hardening procedure that addresses four shortcuts: letter-position bias, distractor leakage, mutual exclusivity and safe-action shortcut, and stem leakage, detailed in the hardening details subsection. We additionally audit whether the radar text itself exposes gold answers to numeric matching (Appendix C.4).

C.2 Final Benchmark Statistics

After hardening and human verification, the final benchmark contains 2,019 multiple-choice QAs distributed across 8 scenario categories and 4 reasoning taxonomies (Table 2). Of the 2,240 candidate QAs generated from the 140 clips, 221 (9.9%) were discarded during verification and 2,019 were retained. Every QA is inspected by human annotators for factual correctness of the gold answer against the source clip, plausibility and mutual exclusivity of the distractors, absence of residual stem leakage, and correctness of the reasoning taxonomy label. Items failing any check are discarded rather than revised, so the discarded fraction accounts for the entire gap between the two counts. Radar availability plays no part in this filtering: it is enforced earlier, at video selection (Appendix D), so all 140 evaluation clips carry at least one front-center radar sensor. Of the 2,019 final QAs, 1,763 (87.3%) are paired with high-resolution imaging long-range radar, while 256 (12.7%) use a short-range fallback radar. See Appendix D for the full sensor-availability breakdown.

C.3 Hardening Details

The hardening procedure addresses four shortcuts that commonly compromise VLM-generated benchmarks. We de-

scribe each below.

Letter-position bias. LLM-generated multiple-choice QAs often place the gold answer at a systematically biased position (e.g., disproportionately on option C). We rebalance the gold-letter distribution toward uniform by permuting options on a per-question basis, ensuring that text-only models cannot exploit position priors.

Distractor leakage. A distractor is *leaky* if a text-only model can identify it as wrong from the question alone (e.g., the distractor is grammatically inconsistent with the question stem, or is far less specific than the gold answer). We identify leaky distractors using a text-only GPT-5-mini probe and rewrite them with re-mined alternatives of comparable plausibility, matched in length and specificity.

Mutual exclusivity and safe-action shortcut. Two failure modes are handled together. First, mutual exclusivity: no two options may be simultaneously true under any plausible interpretation of the scene. Second, the safe-action shortcut: when the gold answer is a safe maneuver (e.g., "slow down"), at least one distractor must also be a plausible safe maneuver, so that a model cannot win by defaulting to safety-oriented commonsense.

Stem leakage. Question stems sometimes contain framing phrases that leak the answer. For example, "*considering adverse weather*, what should the ego vehicle do?" implies that weather is the relevant factor before the question is even read. We remove such leaky framing and rewrite stems to be neutral.

C.4 Numeric-Leakage Audit

Because many gold answers are metric quantities and the radar block is a list of numbers, a model could in principle match the gold option to a radar value without any reasoning. We audit this directly. For each of the 743 QAs whose five options are numeric, we compute for every option the minimum absolute distance to any scalar appearing in that QA's radar text (coordinates, Doppler, intensity, and derived range), and rank the five options by that distance. If numeric matching were a viable shortcut, the gold option would rank first far more often than chance. It does not. The gold option ranks closest in 20.05% of these QAs against a chance level of 20.32% (the options-per-question mean is 4.92 after deduplication), a ratio of $0.99\times$, and the full rank distribution is close to uniform (Table 3). We therefore find no evidence that the radar text exposes gold answers to numeric matching.

Rank of gold option	Share of 743 numeric QAs
1 (closest to a radar value)	20.05%
2	21.67%
3	22.61%
4	18.30%
5 (farthest)	17.36%
Chance (1/mean # options)	20.32%

Table 3: Numeric-leakage audit. Options are ranked by minimum distance to any scalar in the QA's radar text. The gold option is ranked closest at chance level, and the rank distribution is close to uniform.

Reasoning Taxonomy	lrr_1	srr_0	Total
Perception	441	57	498
Anticipation	445	67	512
Reasoning	428	65	493
Planning	449	67	516
Total	**1,763 (87.3%)**	**256 (12.7%)**	**2,019**

Table 4: Radar sensor split across the $2{,}019$ evaluation QAs.

D Radar Sensor Availability

The NVIDIA PhysicalAI-AV dataset provides two front-facing radar sensors per clip: `radar_front_center_imaging_lrr_1` (imaging long-range radar with high angular resolution and dense Doppler returns, our preferred sensor) and `radar_front_center_srr_0` (short-range radar with lower angular density, used as fallback). Our radar-text builder selects the preferred sensor when available and otherwise falls back to the short-range sensor.

Radar availability is enforced at video selection rather than after QA generation: clips lacking both front-center sensors were removed from the candidate pool before the 140 evaluation clips were chosen, and therefore contribute none of the $2{,}240$ generated QAs. For transparency our release still ships the 212 QAs that would have been generated from 15 such clips, marked `excluded=no_front_radar`; they are not part of the benchmark and are excluded from every number reported in this paper. Among the 140 retained clips, 30 lack only the imaging long-range sensor. We retain these because the short-range fallback still provides usable Doppler and range information for evaluation.

Throughout all evaluation conditions (T+V, T+R, T+V+R, and our mmPointRet setting), the radar-text format, point-selection cap, FOV filter, and Doppler ego-motion compensation are applied identically regardless of sensor source.

E mmPointRet LoRA Training Details

The referent-selection component of mmPointRet (Question-Guided Radar Point Retrieval subsection) is implemented as a LoRA adapter applied to Qwen2.5-VL-3B-Instruct. The adapter is trained to take an image with N overlaid candidate bounding boxes (each labeled 1 through N in shuffled order) together with a natural-language question, and output an integer $K \in [1, N]$ identifying the box that corresponds to the question's referent. The vision encoder is kept frozen; only language-model attention projections receive LoRA updates. The trainable adapter contains 7.37M parameters (0.196% of the 3.75B-parameter base) and serializes to $40\,\text{MB}$.

Training data. We construct training pairs from a 500-clip subset of NVIDIA PhysicalAI-Autonomous-Vehicles that is disjoint from the clips used at evaluation. The clip pool was sampled top-150 per category over 8 driving categories ($1{,}051$ candidates), with the 140 edge-case clips

Category	lrr_1	srr_0	Total
Adverse Weather	139	55	194
Aggressive Cut-in	221	24	245
Complex Intersection Crossing	244	1	245
Entry/Exit	262	43	305
Micro-Mobility	168	88	256
Occlusion & Dart-out	273	2	275
Roundabout	222	14	236
Sudden Braking	234	29	263
Total	**1,763**	**256**	**2,019**

Table 5: Sensor split by scenario category.

used for evaluation deduplicated out and an 8-category balanced quota of 500 retained. For each clip, we generated 10 multiple-choice referent questions using the same gpt-5-mini teacher pipeline used to build the main benchmark, yielding 4,991 raw QAs. Following the teacher's per-QA referent track ID, we treat the LiDAR-derived 3D box of that track at the cited frame as the ground-truth referent box; its 8 corners are then projected through the per-clip fisheye intrinsics and rig-to-camera extrinsics to obtain the corresponding 2D box on the image plane. The candidate-bounding-box list in the prompt is the set of all visible 3D LiDAR boxes (strict-FOV filtered) at that frame, similarly projected. Ground-truth K is the post-shuffle position of the referent's projected box within this candidate list.

We then apply three filters to obtain the final training corpus: the teacher emitted a valid referent (4,952/4,991), the question anchors to a single frame (4,951), and at least one LRR radar point lies inside the GT referent OBB after a $\pm 0.05\,\mathrm{m}$ per-axis expansion. The third filter is essential because samples with zero radar in the referent are degenerate for the K-classification objective and additionally bias greedy decoding toward an empty-output token. After all filters, the training corpus contains 1,929 image-question-K triplets drawn from 399 train clips, with 993 held-out validation triplets drawn from a disjoint 100 val clips (clip-level $80/20$ split, seed $= 42$). We stress that this filter is a property of the *training* corpus, where a LiDAR-verified interior return is required for the label to be well defined. It does not characterize inference: at test time the 2D wedge returns a non-empty point set for 98.2% of the 2,019 evaluation QAs (Appendix I.9).

We note a deliberate train-inference asymmetry: training uses LiDAR-GT-derived 2D boxes (perfect localization, no detector noise), whereas inference uses YOLO11 detections. This is an intentional design choice; training mmPointRet against high-quality candidates aligns its target behavior with the upper bound on referent grounding (see Limitations).

LoRA configuration. LoRA updates are applied to the language-model attention projections `q_proj`, `k_proj`, `v_proj`, and `o_proj`, while the vision tower is left untouched. We use rank $r = 16$, scaling factor $\alpha = 32$, dropout 0.05, and no bias. Training is performed in bfloat16 numerical precision with gradient checkpointing enabled.

Training schedule. We optimize the standard next-token cross-entropy loss with all prompt tokens masked (label $= -100$); only the single target digit token is loss-bearing. We use AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$, weight decay $= 0$) with learning rate 2×10^{-4} and a cosine schedule annealed to 0. Warm-up is set to 50 steps. Per-device batch size is 2 and gradient accumulation is 4, yielding an effective batch size of 8. Training runs for 3 epochs, totaling 726 optimizer steps over the 1,929 training triplets. Maximum sequence length is 2048, and image patches are capped at $512 \cdot 28^2$, corresponding to approximately 480 image tokens after 2×2 patch merging. We use random seed 42 for split, bounding-box order shuffle, and sampler.

Training was performed on a single NVIDIA RTX PRO 6000 Blackwell GPU (96 GB VRAM, sm_120) with PyTorch 2.7 (CUDA 12.8) and the `transformers`, `peft`, and `accelerate` stack. Peak training VRAM is approximately 21 GB and wall-clock training time is 41 minutes.

Inference latency. mmPointRet inference on a single QA averages 0.2–0.3 s on the RTX PRO 6000 Blackwell GPU (bfloat16, batch 1): processor preprocessing of a 1920×1080 image plus one forward pass terminating after a single generated token. End-to-end measurements over 200 validation queries yielded 37 s total (after a one-time ≈ 2 s model-load cost), i.e. 185 ms per QA. The downstream wedge projection and radar filtering (Appendix F) are pure NumPy and add less than 5 ms on a single CPU thread, so the full pipeline (excluding the upstream YOLO11 detection step) runs at approximately 200 ms per query at bfloat16. The same adapter loads and runs in ≤ 9 GB of VRAM, enabling deployment on consumer hardware such as RTX 4090 and RTX 3090.

Held-out validation accuracy. On the disjoint 993-triplet validation set, mmPointRet achieves 51.8% top-1 K-classification accuracy, measured on the 110 samples with nonempty GT radar referent. This figure is computed on a small evaluation pool ($n = 110$; Wilson 95% interval $[42.5, 61.0]$) because the nonempty-referent condition is restrictive, and should be read as an indicative rather than a precise estimate. Downstream conclusions in the main paper do not rest on it: the operative comparison is against the question-agnostic wedge control (Appendix I.7), which is measured on the full benchmark.

F Bbox-to-Radar Projection

Given the bounding-box index $\hat{k}$ selected by mmPointRet (Question-Guided Radar Point Retrieval subsection), we deterministically filter the radar point cloud to obtain the subset $\mathbf{P}_Q$ associated with the referent. Let $b_{\hat{k}} \subset [0, W) \times [0, H)$ denote the selected 2D bounding box in pixel coordinates of the front fisheye camera (NVIDIA `camera_front_wide_120fov`, 120° FOV), and let $\pi : \mathbb{R}^3 \to \mathbb{R}^2$ be the dataset's fisheye forward-projection model (a 5-coefficient polynomial parameterised by the manufacturer-provided intrinsic calibration).

Radar set selection. For each radar point $\mathbf{p}_{\text{rig}} \in \mathbb{R}^3$ expressed in the rig frame (after the standard NVIDIA radar-to-rig extrinsic transform), we retain it if its forward projection

falls inside the bounding box:

$$\mathbf{P}_Q \;=\; \big\{\mathbf{p}_{\text{rig}} \;:\; \pi(\mathbf{T}_{\text{rig}\to\text{cam}}\,\mathbf{p}_{\text{rig}}) \in b_{\hat{k}}\big\},$$

where $\mathbf{T}_{\text{rig}\to\text{cam}} \in SE(3)$ is the rig-to-camera extrinsic. Equivalently, $\mathbf{P}_Q$ is the angular wedge in the rig frame whose pixel preimage under π is exactly $b_{\hat{k}}$. Because the fisheye projection π is nonlinear, the wedge boundary is generally curved rather than planar; this curvature is captured exactly by the forward-projection definition above. We additionally enforce $x_{\text{rig}} \geq 0.3\,\text{m}$ to discard radar points behind the camera, matching the FOV pre-filter used during QA construction.

No depth cutoff. The wedge $\mathbf{P}_Q$ is unbounded along the line of sight: radar returns at any range along directions whose pixel projection falls inside $b_{\hat{k}}$ are retained. This is required because our deployment has no LiDAR depth prior to truncate the wedge. In dense scenes this admits points from far-field background objects sharing the referent's bearing. We accept this recall–precision trade-off because the downstream VLM, which receives the full camera frame in parallel, can disambiguate using visual cues that the radar block alone cannot supply. The alternative of bounding the selection with a monocular metric-depth prior is evaluated in the main paper; the construction of that variant is described in Appendix I.4. Appendix I.8 examines the consequence of this trade-off and finds downstream accuracy stable across the wide variation in wedge purity that the untruncated design produces.

Per-frame radar text construction. The retained set $\mathbf{P}_Q$ is serialized in the same Frame-Centric Chronological Block format used by the full-radar baseline (Appendix B.3), with one block per sampled frame listing $(x, y, z, \text{doppler}, \text{intensity})$ for points inside $\mathbf{P}_Q$ at that frame. Unlike the baseline, the point list now contains on average $|\mathbf{P}_Q|$ points per frame instead of the per-frame top-200 cap. Both conditions therefore serialize the same twenty sampled frames and differ only in how many returns each frame contributes. Note that $b_{\hat{k}}$ is selected once, on the frame the question cites, and the same box is applied to every frame of the sequence; no per-frame redetection or tracking is performed. Coverage consequently declines with temporal distance from the cited frame as the referent drifts out of the bearing interval the box subtends, which is quantified in Appendix I.9.

G Master Prompts

This appendix consolidates the prompts used in our pipeline: the master prompt for QA generation (QA generation prompt subsection) and the master prompt for downstream VLM evaluation (evaluation protocol subsection).

G.1 Master Prompt for QA Generation

QAs are generated by `gemini-2.5-flash` with dynamic thinking budget, temperature $= 0.2$, and JPEG quality $= 80$. The model receives 20 video frames sampled at 1 fps and the prompt template below. Slot values are substituted before each call; the template body itself is frozen across all modality conditions for fair ablation. We emphasize that the sensor metadata described below is supplied only to the *generator*, so that gold answers are anchored to measured physical quantities rather than to the generator's own perception. It is never supplied to the models under evaluation (Appendix G.2).

QA Generation Template

```
# System Role
You are a lead AI researcher expert in
     autonomous driving. Your task is
    to construct a highly challenging,
     physical reasoning QA benchmark
    to evaluate the spatial, kinematic
    , and causal understanding of
    multimodal AI models.

# Input Context
- Video File Name: {filename}
- Target Edge Case Category: {
    category_name}
- Category Definition: {
    category_definition}
- Visual Input: 20 sequential video
    frames sampled at 1 fps.
- Provided Sensor & Metadata:
{sensor_metadata}

# Task Instructions
Your goal is to use ALL available
    information provided above to
    generate the most physically
    accurate, high-level, and complex
    multiple-choice questions possible
     regarding this driving scene.

Analyze the visual flow and cross-
    reference it with the [Provided
    Sensor & Metadata]. Focus strictly
     on the "critical moment" of the
    driving scenario. Generate exactly
     4 objective multiple-choice
    questions, one for each taxonomy:
1. Perception: State, exact location,
    depth, relative velocity, or
    occlusion.
2. Anticipation: Trajectory, speed
    changes, or required kinematic
    adjustments (e.g., Time-To-
    Collision, deceleration).
3. Reasoning: The complex physical or
    interactive cause of the critical
    event.
4. Planning: The safest immediate
    action the ego-vehicle must take.

# Generation Rules
1. Concise Options (Anti-Length Bias):
     Each option (A-E) MUST be highly
    concise, maximum 15 words per
    option. Ensure all 5 options are
    roughly the SAME LENGTH so the
    evaluating model cannot guess the
```

```
    answer by length.
2. Direct & Physical Questions (Anti-
    Jargon): Maximum 50 words for the
    question. DO NOT use verbose
    academic jargon or unnecessary
    meteorological/optical
    explanations (e.g., strictly avoid
     terms like "Mie scattering").
    State the practical driving
    problem directly. Maximize the
    physical and kinematic complexity
    strictly based on the provided
    inputs.
3. Deep Rationale: The `rationale` has
     NO length limit. You MUST deeply
    explain the physical reasoning,
    using kinematic logic and exact
    evidence from the provided inputs
    to prove why the answer is correct
    .
4. Temporal Accuracy: Explicitly
    reference specific frame numbers
    briefly in the question (e.g., "At
     frame 15...").
5. Output Format: All text must be in
    English.

# Output Format
Output strictly as a JSON array
    matching this structure. Do not
    output anything else.
[
  {
    "id": "{filename}_Q1",
    "filename": "{filename}",
    "category": "{category_name}",
    "taxonomy": "Perception",
    "question": "[Insert direct,
        physics-focused question. Max
        50 words.]",
    "options": {
      "A": "[Brief Option A. Max 15
          words.]",
      "B": "[Brief Option B. Max 15
          words.]",
      "C": "[Correct Option. Max 15
          words.]",
      "D": "[Brief Option D. Max 15
          words.]",
      "E": "[Brief Option E. Max 15
          words.]"
    },
    "answer": "C",
    "rationale": "[Provide a deep
        physical rationale, explaining
        the exact evidence and logic.
        No length limit.]"
  }
]
```

Concrete example. For a Sudden Braking clip, the slot `{filename}` is set to the clip identifier (e.g., `chunk_1041_7b182c60-...mp4`), `{category_name}` to `Sudden Braking`, and `{category_definition}` to "the gap to the lead vehicle closes rapidly and the ego vehicle decelerates aggressively." The slot `{sensor_metadata}` is filled with the clip's ego-motion, object-track, and LiDAR-box records at the cited frames; a single benchmark is produced from this one generation pass, and the modality conditions of the main paper differ only in what is shown to the evaluated model, never in the questions or gold answers. The visual input consists of 20 JPEG frames at 1 fps from the clip, with Set-of-Mark wireframes drawn on FOV-visible objects.

Generated QA Example

```
{
  "id": "chunk_1041_7b182c60-...mp4_Q5
      ",
  "category": "Sudden Braking",
  "taxonomy": "Anticipation",
  "question": "If the lead vehicle (the
       black sedan) maintained its
      Frame 9 longitudinal acceleration
      , what would its speed be at
      Frame 10?",
  "options": {
    "A": "Approximately 12.5 m/s.",
    "B": "Approximately 8.6 m/s.",
    "C": "Approximately 10.0 m/s.",
    "D": "Approximately 15.3 m/s.",
    "E": "Approximately 11.7 m/s."
  },
  "answer": "B",
  "rationale": "At Frame 9, the lead
      vehicle (the black sedan) has
      speed = 15.26 m/s and a_x = -6.71
       m/s^2. Maintaining this
      acceleration for 1 s gives v(F10)
       = 15.26 + (-6.71)(1) = 8.55 m/s
      ~ 8.6 m/s."
}
```

Note. The exact kinematic constants in this rationale come from the sensor metadata supplied to the generator. They are part of the benchmark's construction record and are not visible to any evaluated model.

G.2 Master Prompt for Evaluation

At evaluation time, every model (Gemini, OpenAI, Qwen, LLaVA) receives an identical prompt body so that scores are directly comparable across providers and modality conditions. The two slots `{question}` and `{options}` are filled per QA; modality-specific channels (radar text, ego-future overlay) are inserted by separate template variants, but the question, options, and output-format sections remain byte-identical. No sensor metadata, gold answer, or generator rationale is present in any evaluation condition.

Evaluation Template

```
# System Role
You are an expert autonomous driving
    AI. You will be shown a front-
    camera driving video (~20 seconds)
     and a multiple-choice question.
    Your task is to select the single
    best answer from the 5 options.

# Visual & Notational Rules (CRITICAL)
1. Frame References: The original
    video was sampled at 1 fps. "Frame
     N" or "F<N>" in the question (e.g
    ., Frame 18) refers to the video
    timestamp at approximately (N - 1)
     seconds (e.g., Frame 18 ~ 17.0 s)
    .
2. Object IDs (Set-of-Mark): The video
     may contain 3D bounding box
    wireframes labeled with numeric
    tags (e.g., [1], [2]). When the
    question references "[K]", "Object
     K", or "ID K", you MUST visually
    locate the bounding box whose
    label is exactly K and analyze
    that object. If no such tag is
    visible, the question is about the
     general scene (no specific object
     overlay was drawn).

# Task
Analyze the scene's kinematics,
    spatial distances, causal
    interactions, and ego-vehicle
    safety. Compare all 5 options and
    pick the single most accurate
    answer.

# Question
{question}

# Options
{options}

# Output Format (MANDATORY)
First, write a BRIEF reasoning (2-4
    sentences) explaining the physical
     and visual evidence that
    justifies your choice. Then, on
    the LAST LINE of your response,
    write EXACTLY one line in this
    format and nothing after it:

Final Answer: X

where X is one of A, B, C, D, E.
```

Inference setting. For OpenAI reasoning models we use `max_completion_tokens=16384`; for Gemini we use the dynamic thinking budget; for non-reasoning models we use temperature $= 0$. The 20 JPEG frames are sent as interleaved "`Frame N:`" text labels and `image_url` parts (`detail="high"`) for OpenAI, and as a native `video/mp4` part for Gemini. An answer is scored correct only if the final line matches the mandated format and names the gold letter; responses without a well-formed final line are scored incorrect. Per-condition rates of such responses are reported in Appendix I.9.

Model Response Example (verbatim log excerpt)

```
The black sedan directly ahead grows
    rapidly in the frame between Frame
     8 and Frame 9 while its rear
    lamps brighten, so the gap is
    closing and the vehicle is braking
     hard rather than coasting. The
    radar block for the selected
    referent shows returns at a
    decreasing range with a strongly
    negative, shrinking Doppler
    magnitude across the same interval
    , which is consistent with a large
     deceleration continuing into
    Frame 10. Extrapolating that trend
     one second forward puts the speed
     well below its Frame 9 value but
    clearly above a full stop, which
    rules out the two highest options
    and the near-stationary one.

Final Answer: B
```

H Additional Ablation: Perturbation Analysis

In addition to the attribute-removal ablation in the Experiments section, we further analyze which aspects of the radar signal the model actually uses by perturbing specific dimensions of the radar input. We perturb a single aspect (frame order, lateral coordinate, or Doppler magnitude) while leaving all other modalities and the rest of the radar intact, and measure the resulting drop in T+V+R accuracy.

H.1 Taxonomy-Aware Subset Construction

Each perturbation is applied only to the subset of QAs whose answer mechanistically depends on the perturbed aspect. Subsets are constructed by an LLM-as-judge classifier (gpt-5-mini) applied to each question's text with a task-specific rubric.

Doppler-shuffle subset (velocity-sensitive, $n = 1{,}171$). A question is included if its correct answer requires reasoning about explicit velocity magnitudes (m/s, km/h), acceleration or deceleration values, closing, relative, or absolute speeds, approach/separation/stationary judgments, or time-to-collision estimates. Representative examples include "what is the approximate average deceleration rate of the ego vehicle from Frame 14 to Frame 16?" and "at Frame 18, what is the longitudinal relative velocity of the pedestrian to the

Perturbation	Disrupts	Full Δ (pp)	Subset Δ (pp)	Subset n
Time-shuffle	Frame order	-5.15	$\mathbf{-7.03}$	1,238
Y-flip	L/R coord.	$+0.79$	-4.32	879
Doppler-shuffle	Velocity mag.	-0.59	-2.48	1,171

Table 6: Effect of radar-targeted perturbations on T+V+R accuracy (%). All Δ values are relative to T+V+R computed on the matching pool (full 2,019 or the indicated subset).

ego?" Questions are excluded if they reference numerical quantities but not velocity (e.g., position-only questions).

Y-flip subset (left/right-sensitive, $n = 879$). A question is included if its correct answer requires reasoning about left/right directional cues (swerve direction, lane choice), lateral position or y-coordinate, adjacent-lane geometry, or objects referenced by lateral position (e.g., "the cyclist on the right"). Representative examples include "what is the optimal lane positioning strategy when passing the rider at Frame 11?" and "after the police car clears the intersection from the left, ..." Questions are excluded if their correct answer is direction-agnostic (e.g., "what is the most prudent action?" or "decelerate firmly").

Time-shuffle subset (interaction-sensitive, $n = 1{,}238$). A question is included if its correct answer requires reasoning about cut-in, dart-out, or crossing dynamics, temporal braking or acceleration sequences of surrounding agents, pedestrian or cyclist trajectory over time, "between Frame X and Frame Y" framing, or past-trajectory-based future prediction. Representative examples include "at Frame 10, how much of the black SUV was occluded by the bus in the preceding frames?" and "what is the ego's approximate average speed change from $t = 0$ s to $t = 3$ s?" Questions are excluded if their correct answer is time-order agnostic (e.g., "safe-driving principles in rain" or "long-term strategy").

Subset overlap. The three subsets are not disjoint: a QA may appear in multiple subsets if its answer depends on multiple radar aspects (e.g., a cut-in question referencing both lateral position and temporal trajectory). Averaged over the 2,019 QAs, each QA appears in 1.6 subsets.

H.2 Perturbation Results

We report perturbation results on both the taxonomy-aware subsets and the full 2,019-QA benchmark. The setup is identical across both: GPT-5-mini, 20 frames at 1 fps, radar text capped at 200 points per frame (top-100 $|v_d|$ $\cup$ top-100 intensity), perturbation applied to the radar text only.

Subset-level hierarchy. On the taxonomy-aware subsets, a clear hierarchy emerges. Time-shuffling produces the largest drop (-7.03 pp), confirming that the model relies on temporal frame ordering for interaction reasoning. Y-flipping causes a moderate drop (-4.32 pp), indicating that models use radar's left/right coordinates but only partially. Doppler-shuffling causes the smallest drop (-2.48 pp), suggesting current VLMs rely on Doppler sign and coarse position more than on fine-grained velocity magnitudes.

Why the full-set effect is smaller. On the full benchmark, each subset-relevant QA is diluted by QAs designed to be insensitive to the perturbed dimension (e.g., direction-agnostic Planning questions for Y-flip), as well as by QAs that nominally reference the dimension but are answerable from vision alone. Hence the full-set Δ values are smaller in magnitude.

Sign reversal on Y-flip. The full-set Y-flip Δ is mildly positive ($+0.79$ pp) rather than zero. On direction-irrelevant QAs the corrupted lateral coordinate acts as mild distractor noise that the model partially discounts, while on the direction-sensitive 879-QA subset the same perturbation degrades accuracy by -4.32 pp. The aggregate yields a small net positive on the full set, confirming that Y-coordinate sensitivity is concentrated in directionally-grounded questions.

Interpretation. We view the subset Δ as the measurement of whether the model uses a given radar aspect for grounding, and the full-set Δ as the measurement of how much that aspect contributes to overall benchmark accuracy. This is actionable: future radar-to-text representations could, for example, discretize Doppler into approach/recede tokens with little expected loss.

I Extended Analyses

This appendix supplements the analyses reported in the Experiments section of the main paper. Appendix I.1 gives the complete results table with the per-scene breakdown referenced from the main text; Appendix I.2 reports the statistical-significance analysis of the mmPointRet gains; and Appendices I.3, I.4, and I.5 document the construction of the retrieval baselines, the depth-truncated variant, and the LiDAR-free retriever whose results are reported in the retriever-comparison tables of the main paper. Appendices I.6–I.9 add four analyses that are referenced from the main text but reported here in full: a scene-level summary of where the gains occur, a decomposition of the retrieval mechanism into referent selection versus prompt compression, a check that the benefit is not confined to the cleanest retrievals, and the retrieved-point and context-cost statistics behind the efficiency claims. The corresponding result tables are not repeated here. An independent non-author review of the benchmark is reported in the Ethical Considerations section.

I.1 Full Results with Per-Scene Breakdown

Table 7 reports the complete evaluation of all nine models under the four input conditions, decomposed by the eight edge-case scenario categories in addition to the four reasoning taxonomies. These are the same runs summarised in the main paper; the per-scene columns are the scene-level results referenced there. Model selection, decoding settings, frame sampling, and the radar text format are identical across all rows, so differences within a model block are attributable to the input condition alone.

Where radar helps most. The largest gains of `ours` over `T+V` concentrate in interaction- and motion-dominated scenes. On Sudden Braking, mmPointRet gives the best accuracy for eight of the nine models, improving over `T+V`

Model	Input	Per-scene accuracy ↑								Per-taxonomy accuracy ↑				Overall ↑
		Wthr	*Cut-in*	*Cross*	*Entry*	*Micro*	*Dart*	*Round*	*Brake*	*Per*	*Ant*	*Rea*	*Pla*	
Closed-source														
GPT-5	T+V	57.22	38.37	55.51	53.11	48.05	50.91	50.00	47.53	39.16	39.65	64.10	57.17	49.98
	T+R_{cap}	49.48	43.67	55.10	56.72	48.83	47.27	51.27	49.43	44.78	51.56	53.75	51.36	50.37
	T+V+R_{cap}	**61.86**	45.71	**60.82**	59.67	**54.69**	55.27	**57.20**	**58.17**	**46.59**	**52.54**	**69.37**	58.14	56.61
	ours	56.70	**47.35**	57.96	**62.62**	**54.69**	**59.64**	56.36	**58.17**	45.98	51.95	68.97	**60.85**	**56.91**
GPT-5-mini	T+V	57.22	40.00	46.12	45.57	48.44	50.55	50.42	44.49	39.16	38.48	58.01	54.65	47.55
	T+R_{cap}	48.97	39.18	44.08	50.16	42.97	45.09	49.58	48.29	38.55	45.90	50.10	49.61	46.06
	T+V+R_{cap}	**60.31**	**44.08**	51.43	54.75	50.00	**52.73**	**53.39**	49.43	41.57	48.24	**60.45**	57.17	51.86
	ours	55.67	40.41	**52.24**	**55.41**	**58.59**	52.36	51.27	**56.27**	**43.57**	**49.22**	59.63	**58.91**	**52.85**
Gemini 2.5 Flash	T+V	51.03	34.69	42.86	50.16	44.92	48.73	47.88	46.18	34.21	30.08	61.46	57.75	45.81
	T+R_{cap}	43.81	35.39	44.90	47.87	44.92	42.91	47.46	43.35	36.14	37.25	50.10	52.13	43.88
	T+V+R_{cap}	54.64	**42.45**	51.03	54.75	53.91	50.18	51.27	**52.09**	40.56	42.35	62.47	**59.88**	51.26
	ours	**58.76**	39.18	**56.33**	**59.02**	**55.86**	**56.00**	**53.39**	50.95	**41.77**	**48.63**	**64.91**	59.69	**53.74**
Open-source														
Qwen3-VL-32B	T+V	**55.15**	**39.18**	47.76	**48.20**	46.09	46.91	46.19	46.77	**38.35**	37.50	55.38	56.20	46.85
	T+R_{cap}	36.60	31.43	38.78	33.44	37.50	34.55	32.63	34.60	23.29	25.59	39.76	50.58	34.87
	T+V+R_{cap}	47.94	38.78	43.67	43.93	44.92	46.91	40.25	42.21	32.53	28.71	56.39	**56.59**	43.54
	ours	47.42	38.37	**49.80**	45.90	**49.22**	**48.00**	**49.58**	**53.61**	36.75	**39.65**	**59.43**	55.23	**47.75**
Qwen3-VL-8B	T+V	**53.61**	**38.37**	35.51	**44.26**	38.67	44.00	41.10	42.97	**35.54**	**28.32**	52.13	52.52	42.10
	T+R_{cap}	42.27	31.43	31.84	28.85	33.59	30.55	36.86	33.08	21.69	18.95	41.78	50.00	33.14
	T+V+R_{cap}	42.27	29.80	39.59	40.33	39.06	42.91	**46.19**	35.74	28.92	22.27	53.35	**53.29**	39.43
	ours	45.36	33.06	**41.63**	39.34	**48.05**	**44.73**	44.07	**44.11**	32.53	28.12	**56.39**	52.91	**42.45**
Qwen2.5-VL-72B	T+V	51.55	37.96	**48.16**	44.26	**53.52**	48.36	41.95	42.97	**37.35**	35.74	**58.62**	52.33	45.96
	T+R_{cap}	44.85	36.33	36.33	37.05	33.98	37.09	40.68	31.94	27.71	31.05	39.55	49.42	37.00
	T+V+R_{cap}	48.97	**39.59**	44.08	44.26	48.44	**48.73**	44.92	42.59	35.14	35.74	57.40	52.33	45.12
	ours	**52.06**	38.78	47.76	**51.15**	51.56	46.18	**47.03**	**47.91**	36.75	**45.31**	55.98	**53.10**	**47.80**
Qwen2.5-VL-32B	T+V	**47.42**	31.84	39.59	44.92	**45.70**	**44.00**	35.17	37.64	32.33	33.20	49.09	48.64	40.81
	T+R_{cap}	36.60	32.65	35.51	33.11	35.55	36.36	35.17	32.70	26.51	27.93	41.78	42.25	34.62
	T+V+R_{cap}	46.39	**37.14**	**43.67**	43.61	40.23	41.09	38.56	34.98	30.92	31.05	50.71	49.81	40.61
	ours	45.88	34.69	42.86	**47.54**	44.14	42.91	**39.83**	**46.39**	**32.93**	**34.57**	**53.96**	**51.16**	**43.14**
LLaVA-Video-7B	T+V	**51.55**	29.80	33.88	**40.66**	**38.67**	**41.09**	**39.83**	32.32	30.32	**29.88**	**48.07**	44.57	38.19
	T+R_{cap}	39.69	33.47	33.06	31.15	32.03	37.82	33.90	32.70	26.31	26.37	38.34	44.96	34.03
	T+V+R_{cap}	45.88	**33.88**	33.47	37.70	33.20	40.00	36.02	36.12	30.32	25.78	43.61	**47.67**	36.85
	ours	47.42	**33.88**	**34.69**	39.67	35.94	40.00	34.75	**42.21**	**32.93**	27.54	46.45	46.90	**38.43**
Qwen2.5-VL-7B	T+V	43.30	29.80	**38.78**	**35.41**	35.94	39.27	35.17	31.94	28.92	27.15	**44.22**	43.80	36.01
	T+R_{cap}	10.31	6.12	5.71	15.41	12.11	15.64	14.41	18.25	9.64	10.94	12.17	17.05	12.48
	T+V+R_{cap}	9.28	5.31	4.08	12.46	8.59	12.73	12.71	18.63	10.64	9.18	10.55	12.21	10.65
	ours	**45.36**	**36.33**	31.43	35.08	**36.33**	**40.73**	**36.44**	**36.50**	**29.52**	**30.66**	44.02	**43.99**	**37.05**

Table 7: **Comparison of radar integration strategies with mmPointRet on mmDrive-QA.** Input conditions: `T+V`, `T+`R_{cap}, `T+V+`R_{cap}, and `ours` (`T+V+`R_{ret}). Scene abbreviations: *Wthr* = Adverse Weather; *Cut-in* = Aggressive Cut-in; *Cross* = Complex Intersection Crossing; *Entry* = Entry/Exit; *Micro* = Micro-Mobility; *Dart* = Occlusion & Dart-out; *Round* = Roundabout; *Brake* = Sudden Braking. Taxonomy abbreviations: *Per* = Perception, *Ant* = Anticipation, *Rea* = Reasoning, *Pla* = Planning. Best result within each model is in **bold**; second-best is underlined.

by $+11.8$ pp for GPT-5-mini ($44.49 \rightarrow 56.27$), $+10.6$ pp for GPT-5 ($47.53 \rightarrow 58.17$), $+9.9$ pp for LLaVA-Video-7B, and $+8.8$ pp for Qwen2.5-VL-32B. Aggressive Cut-in, Entry/Exit, and Occlusion & Dart-out show the same pattern at smaller magnitude (e.g., GPT-5 $+9.0$, $+9.5$, and $+8.7$ pp respectively). These are precisely the categories whose questions turn on the range and closing velocity of one specific agent, which is the quantity the retriever isolates. Appendix I.6 aggregates this pattern across all nine models and all eight categories.

Adverse Weather behaves differently. In Adverse Weather, adding radar in *any* form helps the closed-source models, but the retriever's margin over the per-frame cap is small or slightly negative (GPT-5: 61.86 with R_{cap} vs. 56.70 with R_{ret}), and for Qwen3-VL and LLaVA-Video the vision-only condition remains strongest. Weather questions frequently concern global scene conditions and overall vis-

ibility rather than a single named referent, so restricting the radar block to one object discards context that a broader radar view still provides. A second factor compounds this: the YOLO detector that supplies mmPointRet's candidate boxes runs on the same rain- and low-light-degraded frames, so referent grounding is least reliable exactly where radar would be most valuable. This is the one category where the recall–precision trade-off of question-guided retrieval works against the model, and it points to a natural extension: falling back to a wider radar view when the question has no single referent.

Radar alone is not sufficient. The $\texttt{T+R}_{\texttt{cap}}$ condition is worse than T+V in every scene category for all six open-source models, and mixed for the closed-source ones. Without the image, the model receives metric measurements it cannot attach to a named object, confirming the qualitative analysis in the main paper.

Context load in compact models. The Qwen2.5-VL-7B block makes the token-budget failure explicit at scene granularity: under $\texttt{T+V+R}_{\texttt{cap}}$ its per-scene accuracy collapses to between 4.08% and 18.63%, far below the 20% random-choice floor. The failure is one of output well-formedness rather than of answer quality: 63.1% of responses under this condition contain no well-formed final-answer line, while output truncation accounts for only 0.5% (Table 16). Two controls locate the cause. First, the prompt is not rejected or silently curtailed — it is ingested, and the same prompt is ingested by larger models without incident. Qwen2.5-VL-7B, 32B and 72B share a tokenizer and a context configuration and therefore receive byte-identical inputs averaging $134{,}219$ tokens (Table 15), yet the 32B and 72B models remain functional under exactly this load (40.61% and 45.12%). The collapse is thus capacity-dependent rather than a hard architectural limit: a context load that larger models absorb exhausts the smaller model's ability to hold the output contract while reasoning over the scene. Second, holding the model fixed and shortening only the radar block restores it. With mmPointRet the same 7B model recovers to 31.43–45.36% across all eight scenes and exceeds its own vision-only baseline overall, and its rate of malformed responses returns to the vision-only level of 0.2%. Radar integration at this scale is therefore gated by how much context the radar block consumes, not by whether radar is informative.

I.2 Statistical Significance of mmPointRet Gains

We quantify the reliability of the mmPointRet improvements with paired bootstrap confidence intervals ($n_{\text{boot}} = 10{,}000$, seed $= 42$) and McNemar tests (McNemar 1947) over the full set of 2,019 paired QAs, comparing $\texttt{T+V+R}_{\texttt{cap}}$ against ours (Table 8). Both tests are computed on per-QA correctness indicators from the same runs reported in the main paper; the McNemar statistic uses the exact binomial form on the discordant pairs. Because the $2{,}019$ QAs are drawn from only 140 clips, we resample at the clip level rather than at the QA level, so that all QAs of a clip enter or leave a bootstrap replicate together. Measured design effects lie in $[0.81, 1.25]$, consistent with a near-zero intra-cluster correlation of the paired difference: clip-level factors such as scene difficulty, weather, and detector quality act on both arms of a paired comparison and largely cancel in the difference, even though they correlate strongly with the absolute accuracy of either arm alone. Supplying only the question-relevant radar points yields a statistically significant improvement for the smaller open-source models. For the larger closed-source models (GPT-5, GPT-5-mini), the improvement is not statistically significant, since these models already process large radar prompts without difficulty; even so, mmPointRet matches their accuracy while using $6.4\times$ fewer radar points.

Per-QA correctness logs are available for the five models listed above; for the remaining four, scene-level consistency across the eight scenario categories is summarized instead in Appendix I.6.

I.3 Construction of the Retrieval Baselines

The main paper compares mmPointRet against a LiDAR-oracle retriever and a detector-based wedge filter. Here we specify how each baseline is built. In all cases the downstream VLM (Qwen2.5-VL-72B), the prompt template, the video frames, and the decoding settings are held fixed; *only* the radar block of the prompt is replaced, so any accuracy difference is attributable to the radar-selection strategy alone.

LiDAR oracle. For the QA's annotated referent track, we take the LiDAR-derived 3D box at the cited frame and retain every radar point falling inside that oriented box after the same $\pm 0.05\,\text{m}$ per-axis expansion used to build the training corpus (Appendix E). Because the box supplies both the correct referent identity and its metric extent along the line of sight, this baseline removes the depth ambiguity inherent to the 2D wedge and therefore serves as the performance ceiling for target-aware radar selection. It is an oracle in two senses: it uses the ground-truth referent (no grounding error) and ground-truth depth (no wedge clutter). It is defined on the 707-QA pool of QAs carrying a LiDAR 3D box at the cited frame; for 78 of these the oracle pipeline produced no output and those items are credited with mmPointRet's own correctness. Its accuracy is comparable to the other rows of the retriever table but not to the full-benchmark numbers of the main results table.

YOLO11x + 2D wedge. This baseline isolates the contribution of *question conditioning*. We run YOLO11x on the anchor frame, keep all in-view detections after the same strict-FOV filter, project each detected 2D box into an angular wedge with the identical projection routine of Appendix F, and take the union of the resulting radar subsets. The pipeline is thus identical to ours except that no box is selected: every detected object contributes radar points. Any gap between this baseline and mmPointRet reflects the benefit of selecting the question's referent rather than the benefit of object-level filtering per se. Because it is bearing-anchored and defined on every QA, this control is also evaluated on the full benchmark in Appendix I.7.

Point-quality metrics. Recall, precision, and F1 are computed against the LiDAR-oracle point set as the reference. For each frame we treat the selected and oracle point sets as sets of radar returns identified by their index in the raw

Model	cap (%)	ours (%)	Δ (pp)	95% CI	McNemar p
GPT-5	56.61	56.91	$+0.30$	$[-1.54, +2.08]$	0.792 (ns)
GPT-5-mini	51.86	52.85	$+0.89$	$[-1.09, +2.92]$	0.409 (ns)
Qwen2.5-VL-72B	45.12	47.80	$+2.67$	$[+0.30, +5.05]$	0.0319
Qwen2.5-VL-32B	40.61	43.14	$+2.53$	$[+0.10, +4.86]$	0.0451
Qwen2.5-VL-7B	10.65	37.05	$+26.40$	$[+21.10, +25.91]$	6.3×10^{-67}

Table 8: Paired bootstrap confidence intervals and McNemar tests over 2,019 paired QAs. `cap` denotes $\texttt{T+V+R}_{\texttt{cap}}$ and `ours` denotes $\texttt{T+V+R}_{\texttt{ret}}$; "ns" marks results that are not statistically significant. Accuracies use the strict answer parser applied throughout the paper; under a lenient parser the Qwen2.5-VL-7B cap condition scores 13.67%, still below the 20% chance level, and no conclusion changes.

cloud, and micro-average the counts of true positives, false positives, and false negatives over all frames of all evaluation QAs before computing the three metrics. Precision is low in absolute terms for every 2D method because the oracle set is extremely sparse (4.4 points per frame on average against 508.5 raw returns), so the metrics should be read comparatively rather than as absolute retrieval quality. Point counts in the retriever-comparison table, including the 508.5 raw-return figure, are cited-frame means over the 707-QA pool and are taken from the FOV-filtered scan that feeds the projection stage; the unfiltered scan averages 552 returns per frame. All percentages in that table are formed against the filtered count.

I.4 Depth-Truncated Wedge Variant

To test whether the wedge's unbounded depth (Appendix F) should be truncated, we build a variant that additionally requires agreement between a radar point's range and a monocular metric-depth prediction. We run Depth Anything V2 (Yang et al. 2024) on the anchor frame, take the predicted depth over the selected box $b_{\hat{k}}$, and retain a radar point only if its longitudinal range agrees with that prediction within a tolerance band. Every other component—the referent selected by mmPointRet, the projection routine, the radar text format, and the downstream VLM—is left unchanged.

The variant is far too aggressive: monocular depth is unreliable at the ranges where radar is most informative, so the agreement test removes almost all true target returns, and the retained set collapses to well below one point per frame on average. Precision against the LiDAR oracle rises sharply, as expected from a near-empty selection, but recall and downstream accuracy both degrade substantially (numbers in the main paper). We therefore keep the untruncated wedge and rely on the parallel visual channel for depth disambiguation.

I.5 LiDAR-Free Training (DINORet)

mmPointRet uses LiDAR-derived 3D boxes only as offline supervision during training; at inference the deployed system operates on radar and camera alone, so no LiDAR is required on the vehicle. To show that even the training-time LiDAR requirement can be relaxed, we train *DINORet*, a fully LiDAR-free variant of the retriever.

Supervision source. The only component that changes is how the candidate boxes $\mathcal{B}$ and the target index k^* are obtained. Instead of projecting LiDAR 3D boxes, we run the open-vocabulary detector Grounding DINO-tiny (Liu et al. 2023) on the anchor frame with a text prompt enumerating the driving-relevant object classes, and take its detections as $\mathcal{B}$. The target index k^* is assigned to the detection with the highest overlap against the teacher-annotated referent region; samples with no sufficiently overlapping detection are discarded, as are samples with no radar return inside the selected box, mirroring the filters of Appendix E.

Everything else held fixed. DINORet uses the same Qwen2.5-VL-3B-Instruct backbone, the same LoRA configuration ($r = 16$, $\alpha = 32$, dropout 0.05, attention projections only), the same optimizer and schedule, the same seed, and the same clip-level train/validation split as mmPointRet (Appendix E). At inference it is dropped into the identical pipeline, so the comparison isolates the supervision source.

Outcome. As reported in the main paper, DINORet performs above the vision-only baseline in downstream QA accuracy, confirming that LiDAR-free training is viable. It nevertheless trails LiDAR supervision by roughly 1.5 pp, and the gap is concentrated in recall: vision-based boxes localize the referent less tightly than LiDAR-derived boxes, so a portion of the target's radar returns falls outside the supervised region during training. We therefore adopt the most accurate supervision available in order to establish a strong upper bound, while demonstrating that the LiDAR requirement can be relaxed at a modest cost.

I.6 Scene-Level Summary: Where mmPointRet Helps

Table 9 aggregates the per-scene difference Δ = ours − `T+V` of Table 7 across all nine evaluated models, reporting for each scenario category the mean Δ and the number of models for which it is positive. Because per-QA logs are unavailable for four of the nine models, we report the model-level consistency (how many models agree in sign) rather than a per-QA interval.

Kinematics-dominant scenarios. The gain is largest and most consistent on Sudden Braking: every one of the nine models improves, by $+7.03$ pp on average and by up to $+11.8$ pp (GPT-5-mini). Treating the nine models as independent units, a two-sided sign test on $9/9$ gives $p = 0.0039$; the models span four families (GPT-5, Gemini, Qwen, LLaVA) but are not fully independent, so this should be read as a consistency statistic rather than a benchmark-level test.

Scenario category	Mean Δ (pp)	# positive
Sudden Braking	$+7.03$	**9 / 9**
Micro-Mobility	$+3.82$	6 / 9
Entry/Exit	$+3.24$	5 / 9
Complex Intersection Crossing	$+2.95$	7 / 9
Roundabout	$+2.78$	8 / 9
Aggressive Cut-in	$+2.45$	7 / 9
Occlusion & Dart-out	$+1.86$	6 / 9
Mean over the seven above	$+3.45$	—
Adverse Weather	-1.49	3 / 9

Table 9: Mean Δ (ours $-$ T+V) per scenario category across the nine evaluated models, with the number of models showing a positive difference. mmPointRet improves in seven of the eight categories; Adverse Weather is the single exception.

Aggressive Cut-in, Entry/Exit, Roundabout and Occlusion & Dart-out follow the same pattern at smaller magnitude. These are precisely the categories in which the answer is determined by the range and closing velocity of one specific agent, which is what Doppler radar measures directly and what a monocular camera estimates poorly.

The global-scene exception. Adverse Weather is the only category where the mean is negative (-1.49 pp, positive for three of nine models). Two factors are at work, and neither implicates radar itself. First, weather questions frequently concern global visibility, precipitation, or overall scene risk rather than a single named referent, so a radar block restricted to one object discards context that a broader radar view still supplies. Second, the detector that proposes mmPointRet's candidate boxes operates on the same degraded frames, so referent grounding is least reliable exactly where it is most needed. Radar as a modality continues to help in this category: relative to T+V, the uniform cap improves all three closed-source models by $+3.1$ to $+7.7$ pp (Table 7). What does not transfer is *referent-guided* retrieval. We therefore regard global-scene queries as a scope boundary of the retrieval mechanism rather than a failure of radar-vision-language reasoning, and note the natural remedy: detecting that a question has no single referent and falling back to a wider radar view.

I.7 Selection versus Compression

mmPointRet changes two things at once relative to the uniform cap: it selects a referent, and it shortens the prompt. This appendix separates the two. The natural control is the question-agnostic wedge of Appendix I.3, which is bearing-anchored in exactly the same way as mmPointRet and differs only in that no box is selected. We evaluate it on the full benchmark, where the comparison is adequately powered, using Qwen2.5-VL-72B with every other component held fixed.

Accuracy is non-monotonic in point count. If shortening the prompt were the operative mechanism, accuracy should increase monotonically as the radar block shrinks. It does not (Table 10). Removing 79% of the points by moving from the uniform cap to the question-agnostic wedge ($169.6 \rightarrow 36.0$ points, a $4.7\times$ reduction) changes accuracy by -0.40 pp, an interval that comfortably contains zero. The much smaller further reduction from the wedge to mmPointRet ($36.0 \rightarrow 26.6$, a $1.35\times$ reduction) yields $+3.07$ pp (clip-clustered 95% CI $[+0.84, +5.32]$, McNemar $p = 0.0072$). The two contrasts differ in what changes, not in how much is removed: only the second conditions the selection on the question. Compression alone therefore cannot account for the gain in a model with sufficient context capacity. On the smaller LiDAR-oracle pool of the retriever-comparison table the same contrast is directionally identical but underpowered, which is why we report the full-benchmark comparison as the primary evidence.

Compression still matters, but elsewhere. The above holds for models with the capacity to absorb the uniform cap. For models without it, compression is not a refinement but a precondition: Qwen2.5-VL-7B fails to emit a well-formed answer for 63.1% of QAs under the uniform cap and scores below chance, and recovers fully once the block is compressed (Appendix I.1). Note that the two regimes are separated by model capacity, not by input length: all three Qwen2.5-VL sizes receive the identical 134K-token prompt, and only the smallest fails. The two mechanisms therefore act at different scales — compression makes radar usable for compact models, referent selection makes it useful for large ones — and the main paper's claim rests on both.

I.8 Robustness to Wedge Impurity

The wedge is depth-unbounded by construction (Appendix F), so the retained set contains background returns that share the referent's bearing. Averaged over the 629-QA LiDAR-oracle subset, only 6.0% of retained points (median 3.2%) fall inside the referent's LiDAR box. Retrieval quality therefore varies widely across QAs, and a natural concern is whether the method's benefit is confined to the cleaner cases. We check this: if downstream correctness tracked wedge cleanliness, purity would predict correctness at the QA level.

Regression on retrieval quality. For each QA we compute four measures of retrieval quality against the LiDAR oracle — precision, recall, the raw count of retrieved target points, and a binary indicator of whether any target point was retrieved — and regress per-QA correctness on each, separately for each of the five open-source models with available logs. All twenty regressions return slopes that are not significant at $\alpha = 0.05$ (Table 11 reports the precision slopes; the other three predictors behave identically, with no model reaching $p < 0.05$).

How large an effect can we exclude? A null result is informative only together with the effect size it can rule out. Comparing the top and bottom precision deciles of the 629-QA subset (0.0% versus 14.1% mean precision) gives per-model differences between -14.3 and $+9.5$ pp, all with intervals containing zero, but those intervals are wide (± 16 pp) because each decile holds only 63 QAs. We therefore state the conclusion at the strength the data supports: across four distinct measures of retrieval quality and five models we detect

Radar block	Pts/frame	Acc. (%)	Δ (pp)	Clustered 95% CI
$\mathtt{R_{cap}}$ (uniform, question-agnostic)	169.6	45.12	—	—
YOLO wedge (bearing, question-agnostic)	36.0	44.73	-0.40	$[-2.63, +1.76]$
ours (bearing, question-guided)	26.6	**47.80**	$\mathbf{+3.07}$	$[\mathbf{+0.84}, \mathbf{+5.32}]$

Table 10: Accuracy is non-monotonic in retained point count (Qwen2.5-VL-72B). Accuracies are measured on the full 2,019-QA benchmark; point counts are cited-frame means on the 707-QA pool (Table 12) and are used here only to express the relative size of each reduction. Each Δ is against the row above. Clip-clustered bootstrap; the question-guided contrast has McNemar $p = 0.0072$ and DEFF $= 1.13$.

Model	Slope	SE	p
Qwen2.5-VL-7B	-1.756	1.097	0.109
Qwen2.5-VL-32B	$+0.620$	0.919	0.500
Qwen2.5-VL-72B	$+0.440$	0.902	0.626
Qwen3-VL-8B	$+0.749$	0.918	0.415
Qwen3-VL-32B	$+0.175$	0.900	0.846

Table 11: Logistic regression of per-QA correctness on per-QA retrieval precision, $n = 629$ per model. No slope is significant. Regressions on recall, retrieved-target count, and a binary target-present indicator give the same result for every model.

Convention	Pool	Cap	Ours	Ratio
Cited frame	707	170.48	26.61	6.4×
Cited frame	2,019	165.29	25.50	6.5×
All 20 frames	707	164.68	19.12	8.6×
All 20 frames	629	—	20.40	—
All 20 frames	2,019	162.76	19.56	8.3×

Table 12: Mean retained points per frame under both averaging conventions and each evaluation pool. The retriever-comparison table of the main paper uses the first row (cited frame, $n = 707$); values there are rounded to 169.6 and 26.6.

no relationship between how much target evidence a wedge contains and whether the question is answered correctly, and the analysis excludes only effects substantially larger than the $+3.07$ pp attributable to referent selection (Appendix I.7). The practical reading is that the method does not depend on a clean wedge: its benefit is not confined to the QAs where retrieval happens to be precise, which is what makes the untruncated design usable without a depth prior.

I.9 Retrieved-Point Statistics and Context Cost

Retrieved point sets. Across the full 2,019-QA benchmark, mmPointRet returns a non-empty radar set for 98.2% of questions, with a mean of 19.56 and a median of 14.25 retained points per frame against 508.5 raw returns (Table 13). Only 37 QAs (1.83%) have all twenty frames empty — cases where the selected box projects outside the radar field of view or contains no return at any range — and a further 4 QAs have partial coverage averaging below one point per frame. These fall back to an empty radar block and are answered from vision and text alone.

Two point-count conventions. Retained-point counts in this paper are reported under two conventions that must not be conflated. The *cited-frame* mean averages only over the single frame that the question references, and is the convention used in the retriever-comparison table of the main paper. The *all-frames* mean averages over all twenty sampled frames of each QA, and is the convention that governs prompt cost, since the serialized radar block contains every sampled frame. The two differ because retrieval is most reliable at the cited frame, where the referent is by construction visible and well-localized; subset choice contributes a second, smaller difference (Table 12). Concretely: mmPointRet returns 26.6 points per frame on the cited frame ($n = 707$), and 19.56 points per frame averaged across all twenty frames of each QA ($n = 2{,}019$). Ratios against the uniform cap must be formed within a single convention. The 6.4× reduction quoted in the main paper is the cited-frame ratio on the 707-QA pool and describes how sharply the retriever narrows the scene at the queried moment; the 8.3× all-frames ratio is the one that governs token cost and is the appropriate comparison for Table 14.

Why the two conventions differ for retrieval but not for the cap. Both conditions serialize all twenty sampled frames; the conventions differ only in which frames are averaged when reporting a point count. The gap between them is nonetheless very different for the two strategies. Moving from the cited-frame to the all-frames convention costs the uniform cap 1.5% (165.29 → 162.76 on the full benchmark) but costs retrieval 23.3% (25.50 → 19.56). Expressed per frame, the cap retains 98.4% of its cited-frame count on the remaining nineteen frames, whereas retrieval retains 75.5%. The asymmetry follows from the design. The cap selects scene-wide extremes independently at every frame, so its per-frame count is close to stationary. Retrieval instead selects a single box on the anchor frame and applies that fixed box to the whole sequence (Appendix F), so as the referent and the ego vehicle move, the referent progressively leaves the bearing interval the box subtends and fewer returns fall inside it. Coverage degrades gradually rather than collapsing, because the wedge is depth-unbounded and continues to admit background returns along the same bearing: only 37 QAs (1.83%) end up with all twenty frames empty. This is a known limitation of anchoring on a single frame, and it is the concrete form of the tracking extension noted in the conclusion; propagating the box across frames would raise coverage away from the cited frame at the cost of a tracker dependency.

Avg. points per frame	# QAs	Share
exactly 0 (all 20 frames empty)	37	1.83%
$(0, 1)$	4	0.20%
$[1, 5)$	203	10.05%
$[5, 10)$	411	20.36%
$[10, 20)$	707	35.02%
$[20, 50)$	515	25.51%
$[50, 100)$	115	5.70%
≥ 100	27	1.34%
Total	**2,019**	100.00%

Table 13: Distribution of retained points per frame over the full benchmark, all-frames convention (mean 19.56, median 14.25, max 128.0).

Radar block	Tokens	Pts/frame	Tokens/pt
$\mathtt{R}_{\mathtt{cap}}$ (uniform)	≈108K	162.76	33.2
$\mathtt{R}_{\mathtt{ret}}$ (ours)	≈13.4K	19.56	34.3
Reduction	**8.1×**	**8.3×**	—

Table 14: Cost of the serialized radar block, all-frames convention over the full 2,019-QA benchmark. Token cost is linear in retained point count, so the two reduction factors agree to within 3%.

Radar-block cost. The quantity mmPointRet acts on is the serialized radar block. Measured over the benchmark, it averages ≈108K tokens under the uniform cap and ≈13.4K tokens under question-guided retrieval, an 8.1× reduction (Table 14). This tracks the 8.3× reduction in retained points under the matching all-frames convention: the serialization cost is essentially linear in the number of retained returns, at ≈33 tokens per point in both conditions. Retrieval therefore buys its entire context saving by retaining fewer points, not by encoding them more compactly, and the token ratio can be read directly off the point ratio.

Total prompt length. Table 15 reports measured mean input lengths for the conditions of the main results table. Because Qwen2.5-VL-7B, 32B and 72B share a tokenizer and an image-token budget, these counts are identical across the three model sizes, which is what makes the cross-size comparison in Appendix I.1 a controlled one: the three models differ in capacity while receiving byte-identical input. Absolute counts are not comparable across tokenizer families, since radar text is dominated by floating-point literals that different tokenizers segment very differently; the ratios between conditions are the stable quantity.

Malformed responses. An answer is scored correct only if the response ends with a well-formed final-answer line naming the gold letter (Appendix G.2). Table 16 reports how often that line is absent. For Qwen2.5-VL-7B the uniform cap drives this to 63.1%, which is the immediate cause of its sub-chance score: the model is not selecting badly, it is not selecting at all. Output truncation explains almost none of this (0.5% of responses reach the generation limit), and the rate returns to baseline once the radar block is compressed.

Condition	Mean input tokens
T+V	19,764
$\mathtt{T+R}_{\mathtt{cap}}$	118,485
$\mathtt{T+V+R}_{\mathtt{cap}}$	134,219

Table 15: Measured mean per-QA input length, identical across Qwen2.5-VL-7B, 32B and 72B.

Condition (Qwen2.5-VL-7B)	No final-answer line	Accuracy
T+V	0.2%	36.01%
$\mathtt{T+V+R}_{\mathtt{cap}}$	**63.1%**	10.65%
ours	0.2%	**37.05%**

Table 16: Rate of responses lacking a well-formed final-answer line. The uniform-cap collapse is a formatting failure induced by prompt length, not a degradation of answer quality.

J Representative QA Examples Across Eight Scene Categories

To give a concrete sense of the benchmark, we show eight QA examples drawn from mmDrive-QA: one for each of the eight edge-case scene categories, distributed so that every reasoning category (Perception, Anticipation, Reasoning, Planning) contributes two examples. Clip UUIDs will be released with our QA annotations; the corresponding video clips can be obtained directly from NVIDIA under its license terms (Appendix B.1). The correct option is marked with ★. Object names use natural-language descriptors (e.g., "the silver hatchback" or "the pedestrian").

Example 1: Perception, Adverse Weather

Clip: `chunk_0580_2f020804_....mp4`



Q. At Frame 17, what is the magnitude of the relative velocity of the vehicle with respect to the ego vehicle?

A. Approximately 2.05 m/s.
B. Approximately 0.33 m/s.
C. Approximately 11.51 m/s.
D. Approximately 1.17 m/s. ★
E. Approximately 11.18 m/s.

Rationale. $\mathbf{v}_{\text{ego}} = (-4.14, 10.37, -0.39)$, $\mathbf{v}_{\text{obj}} = (-3.21, 11.06, -0.21)$. $\|\mathbf{v}_{\text{obj}} - \mathbf{v}_{\text{ego}}\| = \sqrt{0.93^2 + 0.69^2 + 0.18^2} \approx 1.17$ m/s.

Example 2: Perception, Roundabout

Clip: `chunk_2833_7f085d8b_....mp4`



Q. At Frame 16, how does the ego vehicle's speed compare to the heavy truck directly ahead?

A. Ego is slightly faster.
B. Ego is significantly faster.
C. Ego is slightly slower.
D. Ego is significantly slower. ★
E. They are moving at roughly the same speed.

Rationale. Ego speed = 12.83 m/s vs. truck speed = 18.55 m/s; a 5.72 m/s gap is a significant difference.

Example 3: Anticipation, Aggressive Cut-in

Clip: `chunk_2087_2a62d544_....mp4`



Q. At Frame 17, how much time until the white car's center crosses the ego vehicle's left lane boundary (y=0 m), assuming constant lateral velocity?

A. Approximately 4.0 seconds.
B. Approximately 10.0 seconds.
C. Approximately 2.0 seconds.
D. Approximately 6.4 seconds. ★
E. Approximately 8.0 seconds.

Rationale. White car at $y = -2.43$ m with $v_y = 0.38$ m/s. Crossing distance $= 2.43$ m; $t = 2.43/0.38 \approx 6.4$ s.

Example 4: Anticipation, Sudden Braking

Clip: `chunk_3101_69f0493f_....mp4`



Q. At Frame 17, what is the predicted spatial relationship between the ego's path and the silver hatchback?

A. The silver hatchback is stationary.
B. Ego passes ahead of the silver hatchback.
C. Paths are parallel.
D. Paths are predicted to intersect. ★
E. Ego passes behind the silver hatchback.

Rationale. Ego is yawing left ($\psi = 0.46$ rad) into the oncoming hatchback's lane. With a closing speed of $\sim$16.0 m/s and TTC ≈ 0.4 s, paths intersect imminently absent evasive action.

Example 5: Reasoning, Occlusion & Dart-out

Clip: `chunk_0172_c778a7a8_....mp4`



Q. Based on the ego's predicted trajectory and the pedestrian's velocity at Frame 20, what is the collision likelihood?

A. High, little time for avoidance.
B. Moderate, ego must brake to yield.
C. Low, their paths are diverging. ★
D. High, crossing into ego left turn.
E. Moderate, brief crossing plausible.

Rationale. Ego at $(67.69, -5.80)$ with $\mathbf{v} = (-0.20, 1.79)$; pedestrian at $(71.12, 10.33)$ with $\mathbf{v} = (1.12, -1.33)$. The pedestrian is ahead and to the right of the ego's left-turning arc, and the two velocity vectors diverge.

Example 6: Reasoning, Complex Intersection Crossing

Clip: `chunk_1878_3cc9184c_....mp4`



Q. What environmental factor, besides the truck, contributes to the complexity of the scene in Frame 2?

A. Confusing temporary traffic signage and cones.
B. Cyclists and scooters present near the intersection.
C. Parked vehicles along the curb blocking sightlines.
D. High density of other vehicles and pedestrians. ★
E. Faded lane markings making lane boundaries ambiguous.

Rationale. The frame shows a busy intersection: many automobiles and several pedestrians populate the field of view alongside the heavy truck, substantially elevating the dynamic-interaction load.

Example 7: Planning, Entry/Exit

Clip: `chunk_2395_72cb4b63_....mp4`



Q. At Frame 13, what is the most appropriate immediate action for the ego vehicle?

A. Decelerate and/or adjust path to avoid conflict. ★
B. Hold lane and maintain current speed given the silver SUV's proximity.

C. Signal and initiate a controlled left lane change at steady speed.

D. Alert the silver SUV (horn/flash) while holding lane and speed.

E. Position toward the right lane edge to increase lateral buffer without braking.

Rationale. The silver SUV's predicted 3-s ego footprint shows substantial overlap with the ego's path; reducing speed or laterally adjusting the trajectory is the only conflict-avoiding option among five otherwise plausible safe actions.

Example 8: Planning, Micro-Mobility

Clip: `chunk_0136_aecdc5a4_....mp4`



Q. At Frame 18, what is the most critical immediate action for the ego vehicle?

A. Begin controlled slowing now.

B. Reduce speed to a smooth stop before the crosswalk.

C. Slow gently to allow yielding by the pedestrian.

D. Decelerate to cautionary speed.

E. Initiate emergency braking. ★

Rationale. Ego at $(32.88, 6.82)$ with $v = 5.75$ m/s and accelerating; pedestrian at $(42.64, 13.07)$ crossing into the ego's path with $v = 1.22$ m/s. TTC ≈ 1.9 s; only emergency braking is fast enough to avert contact.

K Subset Glossary

Several analyses in this paper are necessarily computed on subsets of the benchmark, because they require auxiliary annotations that are not available for every QA. Table 17 defines each one. Accuracies computed on different subsets are not directly comparable; in particular, the retriever-comparison table is defined only on the 707-QA pool and its accuracies should not be read against the full-benchmark numbers of the main results table.

Size	Name	Definition
2,019	Full benchmark	All human-verified QAs.
1,763	LRR subset	Clips with the imaging long-range radar.
1,238	Time-sensitive	Perturbation subset, App. H.1.
1,171	Velocity-sensitive	Perturbation subset, App. H.1.
993	Retriever val.	Held-out K-classification triplets, App. E.
879	Direction-sensitive	Perturbation subset, App. H.1.
743	Numeric QAs	Five numeric options; leakage audit.
707	LiDAR-box pool	LiDAR 3D box at the cited frame.
629	Nonempty GT points	Above, with ≥ 1 radar return inside that box.
110	Nonempty-ref. val.	Validation triplets with a nonempty GT referent.

Table 17: Evaluation subsets used in this paper.

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